

COMANDOS Y PROCEDIMIENTOS

Parte 0: PREPARACIÓN DEL ENTORNO LOCAL.

1. Verificar WSL2.

wsl --version

```
moises@A6Alumno04:~$ wsl --version
Unknown command: --version
WSL
Wslman Shell commandLine, version 0.2.1

USAGE: wsl COMMAND [PARAMS...]

COMMANDS:
identify  - WS-Identify
enum      - WS-Enumerate
get       - WS-Get
put       - WS-Put
invoke    - WS-Invoke
xclean    - Delete all files generated by this tool set
xcred     - Create or display credential file
xcert     - Get server certificate (saved to <IPADDRESS>.crt)

PARAMS specification is specific to a COMMAND.

Output will be saved to ./response.xml. If you want to run parallel
executions in the same directory, define RTFILEPREFIX in the environment
Doing so may significantly increase files generated.

Requires: curl, xmllint, GNU core utilities.
Optional: xsltproc for output formatting, gpg for encrypted credential.
Optional: wget as alternate for curl when not available.
```

2. Crear directorio para las claves SSH.

mkdir -p ~/.ssh

```
moises@A6Alumno04:~$ mkdir -p ~/.ssh
```

chmod 700 ~/.ssh

```
moises@A6Alumno04:~$ chmod 700 ~/.ssh
```

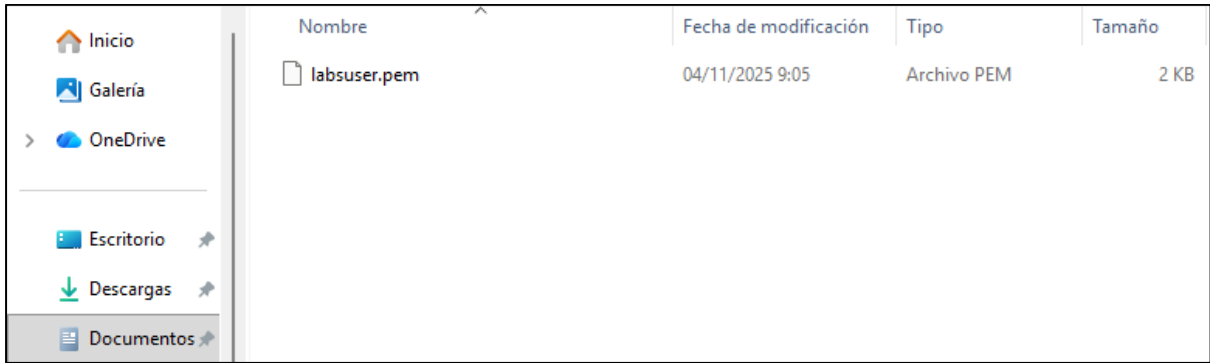
PARTE 1. CONFIGURACIÓN EN AWS (*Interfaz Visual*)

1. Descargar la clave PEM desde la página del laboratorio.
 1. Localiza el botón o enlace **“Download key”** o **“Download .pem”**.

The screenshot displays the AWS Cloud Labs interface. At the top, there is a navigation bar with links: [AWS](#), [Start Lab](#), [End Lab](#), [AWS Details](#), [Readme](#), [Reset](#), and a close icon. Below this, a status bar shows 'Used \$0.2 of \$50' and a timer '03:59'. The main area is split into two panels. The left panel is a terminal window with the prompt 'eee_w_5317559@runweb196019:~\$' and a large '59x47' watermark. The right panel, titled 'Cloud Access', contains several sections: 'AWS CLI' with a 'Show' button; 'Cloud Labs' with session details (Remaining session time: 03:58:24(239 minutes), Session started at: 2025-11-13T05:24:14-0800, Session to end at: 2025-11-13T09:24:14-0800, Accumulated lab time: 00:22:00 (22 minutes), and No running instance); 'SSH key' with a 'Show' button; 'Download PEM' (highlighted with a red rectangle); 'Download PPK'; 'AWS SSO' with a 'Download URL' button; and a table with AWSAccountid (20524) and Region (us-east-1).

Cloud Access	
AWS CLI: Show	
Cloud Labs	
Remaining session time: 03:58:24(239 minutes)	
Session started at: 2025-11-13T05:24:14-0800	
Session to end at: 2025-11-13T09:24:14-0800	
Accumulated lab time: 00:22:00 (22 minutes)	
No running instance	
SSH key Show	
Download PEM	
Download PPK	
AWS SSO	
Download URL	
AWSAccountid	20524
Region	us-east-1

2. Se descargará un archivo con extensión **.pem** (por ejemplo: **labsuser.pem**).



Nombre	Fecha de modificación	Tipo	Tamaño
labsuser.pem	04/11/2025 9:05	Archivo PEM	2 KB

3. **Importante:** guarda este archivo en una ubicación segura, preferiblemente en WSL.

Pasos para mover la clave a WSL:

Si descargaste el archivo en Windows:

```
cp /mnt/c/Users/TU-USUARIO/Downloads/labsuser.pem ~/.ssh/
```

```
moises@A6Alumno04:~$ cp /mnt/c/Users/Alumno.DESKTOP-DI5KTUG/Documents/labsuser.pem ~/.ssh/
```

2. Configurar permisos de la clave PEM.

```
chmod 400 ~/.ssh/labsuser.pem
```

```
moises@A6Alumno04:~$ chmod 400 ~/.ssh/labsuser.pem
```

Verifica los permisos:

```
ls -la ~/.ssh/labsuser.pem
```

```
moises@A6Alumno04:~$ ls -la ~/.ssh/labsuser.pem
-r----- 1 moises moises 1678 Nov 18 09:04 /home/moises/.ssh/labsuser.pem
```

3. Crear instancia EC2

1. Nombre: **servidor-web-practica**
2. AMI: **Ubuntu 22.04 LTS o Ubuntu 24.04 LTS**
3. Tipo: **t2/3.micro (Free Tier)**
4. Red: **VPC y Subred por defecto**
5. IP pública: **Habilitada**
6. Almacenamiento: **8 GiB, tipo gp2**
7. Etiquetas (opcional): **Curso → ArquitecturaNube**
8. Grupo de Seguridad: **sg-servidores-web**

The screenshot displays the AWS Management Console interface for an EC2 instance. At the top, the instance ID 'i-04215907f6d9c8e93' and name 'mi-servidor-web' are shown. Below this is a navigation bar with tabs for 'Estado y alarmas', 'Monitoreo', 'Seguridad', 'Redes', and 'Almacenamiento'. The 'Seguridad' tab is selected, showing 'Detalles de seguridad'. This section includes the 'Rol de IAM' (empty), 'ID del propietario' (205248415840), and 'Hora de lanzamiento' (Thu Nov 13 2025 14:24:11 GMT+0100). A red arrow points to the 'Grupos de seguridad' section, which lists 'sg-0f9a14d78653358e9 (launch-wizard-1)'. Below this, the 'Instancias (1/1)' section is visible, featuring buttons for 'Conectar', 'Estado de la instancia', 'Acciones', and 'Lanzar instancias'. A search bar and a filter 'Estado de la instancia = running' are also present. A red arrow points to the instance list table, which contains one entry: 'mi-servidor-web' with ID 'i-04215907f6d9c8e93', status 'En ejecución', and type 't3.micro'. The table also shows a progress indicator '3/3 completados'.

✓	Name	ID de la instancia	Estado de la i...	Tipo de inst...	Comprobaci
✓	mi-servidor-web	i-04215907f6d9c8e93	En ejecución	t3.micro	3/3 completados

EC2

Panel

Vista global de EC2

Eventos

▼ Instancias

Instancias

Tipos de instancia

Plantillas de lanzamiento

Solicitudes de spot

Savings Plans

Instancias reservadas

Alojamientos dedicados

Reservas de capacidad

Capacity Manager

▼ Imágenes

AMI

Catálogo de AMI

▼ Elastic Block Store

Volúmenes

Instantáneas

Administrador del ciclo de vida

▼ Red y seguridad

Security Groups

Direcciones IP elásticas

Grupos de ubicación

Pares de claves

Interfaces de red

Puede cambiar la página de inicio predeterminada para EC2.

Descartar permanentemente

Cambiar la página de inicio

Recursos

Vista global de EC2

Actualmente, utiliza los siguientes recursos de Amazon EC2 en la región Estados Unidos (Norte de Virginia):

Instancias (en ejecución)	1	Balancedadores de carga	0	Capacity Reservations	0
Direcciones IP elásticas	0	Grupos de escalamiento automático	0	Grupos de seguridad	2
Grupos de ubicación	0	Hosts dedicados	0	Instancias	1
Instantáneas	0	Pares de claves	1	Volúmenes	1

Lanzar la instancia

Para comenzar, lance una instancia de Amazon EC2, que es un servidor virtual en la nube.

Lanzar la instancia

Migrar un servidor

Nota: Sus instancias se lanzarán en la región Estados Unidos (Norte de Virginia)

Alarmas de instancia

Ver en CloudWatch

0 en alarma 0 Aceptar 0 datos insuficientes

Instancias en alarma

Estado del servicio

Panel de AWS Health

Región

Estados Unidos (Norte de Virginia)

Estado

Este servicio funciona con normalidad.

Zonas

Nombre de la zona	ID de la zona
us-east-1a	use1-az2
us-east-1b	use1-az4
us-east-1c	use1-az6
us-east-1d	use1-az1

6

Página de inicio de la Consola Información

Restablecer al diseño predeterminado + Agregar widgets

Visitados recientemente Información

 [EC2](#)

Ver todos los servicios

Aplicaciones (0) Información [Crear aplicación](#)

Región: US East (N. Virginia)

Seleccionar región
us-east-1 (Región actual) [Buscar aplicaciones](#)

Nombre	Descripción	Región
No hay aplicaciones Comience creando una aplicación.		

[Crear aplicación](#)

Ir a myApplications

Le damos la bienvenida a AWS

[Introducción a AWS](#)

Conozca los aspectos fundamentales y encuentre información valiosa para sacar el máximo provecho de AWS.

[Formación y](#)

AWS Health Información

Problemas abiertos
0 Últimos 7 días

Cambios programados
1 Próximos 7 días y últimos 7 días

Otras notificaciones
0 Últimos 7 días

Costo y uso Información

Mes actual
0,18 US\$

Final de mes previsto
0,43 US\$

Oportunidades de ahorro
Acceso denegado

Coste (US\$)

Activar Windows
Ve a Configuración para activar Windows

Nombre y etiquetas Información

Nombre
 [Agregar etiquetas adicionales](#)

Imágenes de aplicaciones y sistemas operativos (Imagen de máquina de Amazon) Información

Una AMI posee el sistema operativo, el servidor de aplicaciones y las aplicaciones de la instancia. Si a continuación no ve una AMI adecuada, utilice el campo de búsqueda o elija [Buscar más AMI](#).

Recientes [Inicio rápido](#)

Amazon Linux


macOS


Ubuntu


Windows


Red Hat


SUSE Linux


Debian


Imágenes de máquina de Amazon (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-0ecb62995f68bb549 (64 bits (x86)) / ami-01b9f1e7dc427266e (64 bits (Arm))
Virtualización: hvm Activado para ENA: true Tipo de dispositivo raíz: ebs [Apto para la capa gratuita](#)

Descripción

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Arquitectura	ID de AMI	Fecha de publicación	Nombre de usuario
64 bits (x86)	ami-0ecb62995f68bb549	2025-10-22	ubuntu

[Proveedor verificado](#)

sg-0f9a14d78653358e9 - launch-wizard-1

Acciones

Detalles

Nombre del grupo de seguridad

launch-wizard-1

ID del grupo de seguridad

sg-0f9a14d78653358e9

Descripción

launch-wizard-1 creada 2025-11-04T08:09:19.597Z

ID de la VPC

vpc-0f583dcf81a9ded72

Propietario

205248415840

Número de reglas de entrada

5 Entradas de permisos

Número de reglas de salida

1 Entrada de permiso

Reglas de entrada

Reglas de salida

Compartiendo : *novedad*

Asociaciones de

Reglas de entrada (5)

Administrar etiquetas

Editar reglas de entrada

Buscar

1

	Name	ID de la regla del gr...	Versión de IP	Tipo
	-	sgr-0b63c6f869a4c2799	IPv4	TCP personalizado
	-	sgr-0cba4936876614ebe	IPv4	TCP personalizado
	-	sgr-00ee5efdb66eb61fe	IPv4	SSH
	-	sgr-007ff8988abb62861	IPv4	TCP personalizado
	-	sgr-09f4ebcf2e48fde66	IPv4	TCP personalizado

4. Configurar Security Group (Reglas de entrada).

Puerto	Protocolo	Tipo	Descripción
22	TCP	SSH	Acceso SSH remoto
8080	TCP	HTTP personalizado	Apache HTTP
8081	TCP	HTTP personalizado	Nginx
8082	TCP	HTTP personalizado	Caddy
8443	TCP	HTTPS personalizado	Apache HTTPS (SSL)

Regla de entrada 5 Eliminar

ID de la regla del grupo de seguridad: sgr-09f4ebcf2e48fde66

Tipo: Información TCP personalizado

Protocolo: Información TCP

Intervalo de puertos: Información 8081

Tipo de origen: Información Anywhere-IPv4

Origen: Información 0.0.0.0/0

Descripción: opcional Información

Agregar regla

Cancelar Previsualizar los cambios Guardar reglas

Nombre	ID de la regla del grupo d...	Intervalo de pu...	Protocolo	Origen	Grupos de seguridad	Descripción
-	sgr-0b63c6f869a4c2799	8082	TCP	0.0.0.0/0	launch-wizard-1	-
-	sgr-0c8a4936876614ebe	8443	TCP	0.0.0.0/0	launch-wizard-1	-
-	sgr-00ee5efdb66eb61fe	22	TCP	0.0.0.0/0	launch-wizard-1	-
-	sgr-007ff8988abb62861	8080	TCP	0.0.0.0/0	launch-wizard-1	-
-	sgr-09f4ebcf2e48fde66	8081	TCP	0.0.0.0/0	launch-wizard-1	-

5. Especificar clave PEM al lanzar la instancia.

Selecciona **Use Existing Key Pair** y elige la clave **labsuser.pem**.

Si la instancia fue creada sin clave, termina la instancia y crea una nueva seleccionando la clave correcta.

PARTE 2: CONEXIÓN SSH DESDE WSL A AWS

1. Obtener la IP pública de la instancia.

En AWS Console:

EC2 → Instances → Public IPv4 address.

2. Conectar por SSH

ssh -i ~/.ssh/labsuser.pem ubuntu@TU-IP-PUBLICA

Ejemplo:

ssh -i ~/.ssh/labsuser.pem ubuntu@54.123.45.67

```
moises@A6Alumno04:/mnt/c/WINDOWS/system32$ sudo ssh -i ~/.ssh/labsuser.pem ubuntu@98.81.164.4
[sudo] password for moises:
The authenticity of host '98.81.164.4 (98.81.164.4)' can't be established.
ED25519 key fingerprint is SHA256:QZUSzZYYGxo6+ENXa0pkqpAdW5ZxufDu5cF5zUMjQKU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.81.164.4' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1016-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Thu Nov 20 13:20:17 UTC 2025

System load:  0.08           Temperature:   -273.1 C
Usage of /:   34.8% of 6.71GB Processes:      108
Memory usage: 23%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.23.219

Expanded Security Maintenance for Applications is not enabled.

4 updates can be applied immediately.
4 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Nov 13 13:30:31 2025 from 18.206.107.29
ubuntu@ip-172-31-23-219:~$
```

PARTE 3: EJECUTAR LA SEGUNDA PRÁCTICA ORIGINAL

A partir de aquí lo tienes todo configurado.

Ejecuta todos los comandos de la Segunda Práctica directamente en la instancia EC2 conectada por SSH.

Accede desde tu navegador local:

<http://TU-IP-PUBLICA:PUERTO>

```
ubuntu@ip-172-31-23-219:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1588 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [300 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.7 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1500 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [304 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.4 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2251 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [510 kB]
```

Después de actualizar los paquetes se instala apache:

```
ubuntu@ip-172-31-23-219:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
```

Se cambia el puerto 80 al 8080 de apache en el fichero de configuración:

```
GNU nano 7.2 /etc/apache2/ports.conf
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 8080

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>
```

Se cambia el host virtual:

```
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf
<VirtualHost *:8080>
# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
```

Instalación y comprobación de estado de PHP:

```
ubuntu@ip-172-31-23-219:~$ sudo apt install php libapache2-mod-php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php8.3 php8.3-cli php8.3-common php8.3-opcache
  php8.3-readline
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.

ubuntu@ip-172-31-23-219:~$ sudo systemctl restart apache2
ubuntu@ip-172-31-23-219:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-20 13:32:44 UTC; 11s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 13398 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 13402 (apache2)
    Tasks: 6 (limit: 1017)
   Memory: 10.5M (peak: 10.9M)
      CPU: 53ms
   CGroup: /system.slice/apache2.service
           └─13402 /usr/sbin/apache2 -k start
             └─13404 /usr/sbin/apache2 -k start
               └─13405 /usr/sbin/apache2 -k start
                 └─13406 /usr/sbin/apache2 -k start
                   └─13407 /usr/sbin/apache2 -k start
                     └─13408 /usr/sbin/apache2 -k start

Nov 20 13:32:44 ip-172-31-23-219 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 20 13:32:44 ip-172-31-23-219 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

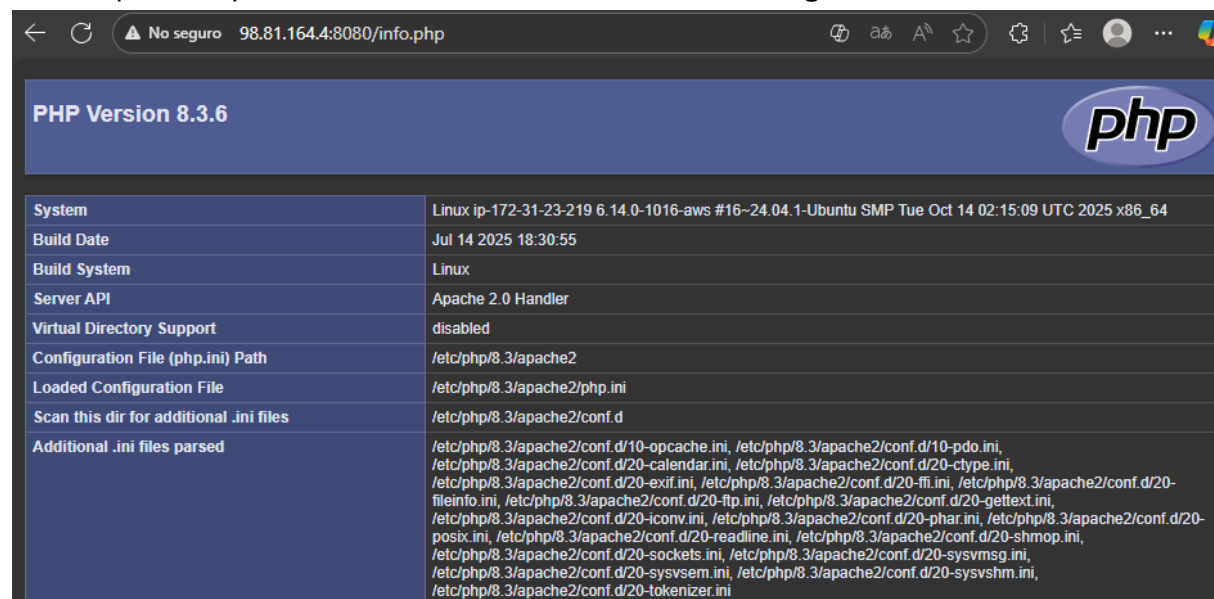
Se comprueba el puerto por el que escucha apache:

```
ubuntu@ip-172-31-23-219:~$ sudo netstat -tulpn | grep apache2
tcp6      0      0 :::8080          :::*              LISTEN    13402/apache2
```

Se añade la página oficial de php:

```
ubuntu@ip-172-31-23-219:~$ echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/info.php
<?php phpinfo(); ?>
ubuntu@ip-172-31-23-219:~$ curl http://localhost:8080/info.php
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml"><head>
<style type="text/css">
body {background-color: #fff; color: #222; font-family: sans-serif;}
pre {margin: 0; font-family: monospace;}
a:link {color: #009; text-decoration: none; background-color: #fff;}
a:hover {text-decoration: underline;}
table {border-collapse: collapse; border: 0; width: 934px; box-shadow: 1px 2px 3px rgba(0, 0, 0, 0.2);}
.center {text-align: center;}
.center table {margin: 1em auto; text-align: left;}
.center th {text-align: center !important;}
td, th {border: 1px solid #666; font-size: 75%; vertical-align: baseline; padding: 4px 5px;}
th {position: sticky; top: 0; background: inherit;}
h1 {font-size: 150%;}
h2 {font-size: 125%;}
h2 a:link, h2 a:visited {color: inherit; background: inherit;}
.p {text-align: left;}
```

Se comprueba que muestra el contenido a través del navegador:



PHP Version 8.3.6

System	Linux ip-172-31-23-219 6.14.0-1016-aws #16~24.04.1-Ubuntu SMP Tue Oct 14 02:15:09 UTC 2025 x86_64
Build Date	Jul 14 2025 18:30:55
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/apache2
Loaded Configuration File	/etc/php/8.3/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/apache2/conf.d
Additional .ini files parsed	/etc/php/8.3/apache2/conf.d/10-opcache.ini, /etc/php/8.3/apache2/conf.d/10-pdo.ini, /etc/php/8.3/apache2/conf.d/20-calendar.ini, /etc/php/8.3/apache2/conf.d/20-ctype.ini, /etc/php/8.3/apache2/conf.d/20-exif.ini, /etc/php/8.3/apache2/conf.d/20-fi.ini, /etc/php/8.3/apache2/conf.d/20-fileinfo.ini, /etc/php/8.3/apache2/conf.d/20-ftp.ini, /etc/php/8.3/apache2/conf.d/20-gettext.ini, /etc/php/8.3/apache2/conf.d/20-iconv.ini, /etc/php/8.3/apache2/conf.d/20-phar.ini, /etc/php/8.3/apache2/conf.d/20-posix.ini, /etc/php/8.3/apache2/conf.d/20-readline.ini, /etc/php/8.3/apache2/conf.d/20-shmop.ini, /etc/php/8.3/apache2/conf.d/20-sockets.ini, /etc/php/8.3/apache2/conf.d/20-sysvmsg.ini, /etc/php/8.3/apache2/conf.d/20-sysvsem.ini, /etc/php/8.3/apache2/conf.d/20-sysvshm.ini, /etc/php/8.3/apache2/conf.d/20-tokenizer.ini

PARTE 2: INSTALACIÓN Y CONFIGURACIÓN DE NGINX

1. Instalar Nginx

Comando:

```
sudo apt install nginx -y
```

Descripción: Instala el servidor web Nginx en tu sistema.

```
ubuntu@ip-172-31-23-219:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 564 kB of archives.
```

2. Configurar Nginx en puerto 8081

Comando:

```
sudo nano /etc/nginx/sites-available/default
```

```
server {
    listen 8081 default_server;
    listen [::]:80 default_server;
```

3. Crear página HTML personalizada

Comando:

```
echo "<h1>Servidor Nginx</h1><p>Funcionando en puerto 8081</p>" | sudo tee
/usr/share/nginx/html/index.html
```

Descripción: Crea una página HTML identificable para Nginx.

```
ubuntu@ip-172-31-23-219:~$ sudo nano /etc/nginx/sites-available/default
ubuntu@ip-172-31-23-219:~$ echo "<h1>Servidor NGINX</h1><p>Funcionando en puerto 8081</p>" | sudo tee /usr/share/nginx/html/index.html
<h1>Servidor NGINX</h1><p>Funcionando en puerto 8081</p>
ubuntu@ip-172-31-23-219:~$ sudo systemctl restart nginx
```

5. Verificar estado de Nginx

Comando:

```
sudo systemctl status nginx
```

Descripción: Comprueba que Nginx está funcionando correctamente.

Comando adicional para verificar puerto:

```
sudo netstat -tulpn | grep nginx
```

Descripción: Verifica en qué puerto está escuchando Nginx (*debe mostrar 8081*).

```
ubuntu@ip-172-31-23-219:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-20 13:41:51 UTC; 30s ago
     Docs: man:nginx(8)
   Process: 13930 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 13934 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 13935 (nginx)
      Tasks: 3 (limit: 1017)
    Memory: 2.4M (peak: 2.6M)
       CPU: 17ms
    CGroup: /system.slice/nginx.service
            └─13935 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─13936 "nginx: worker process"
                └─13937 "nginx: worker process"

Nov 20 13:41:51 ip-172-31-23-219 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server:
Nov 20 13:41:51 ip-172-31-23-219 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server:

ubuntu@ip-172-31-23-219:~$ sudo netstat -tulpn | grep nginx
tcp        0      0 0.0.0.0:8081          0.0.0.0:*            LISTEN      13935/nginx: master
tcp6       0      0 :::80               :::*                  LISTEN      13935/nginx: master
```


6. Probar Nginx desde terminal

Comando:

```
curl http://localhost:8081
```

Descripción: Verifica que Nginx sirve correctamente el contenido HTML.

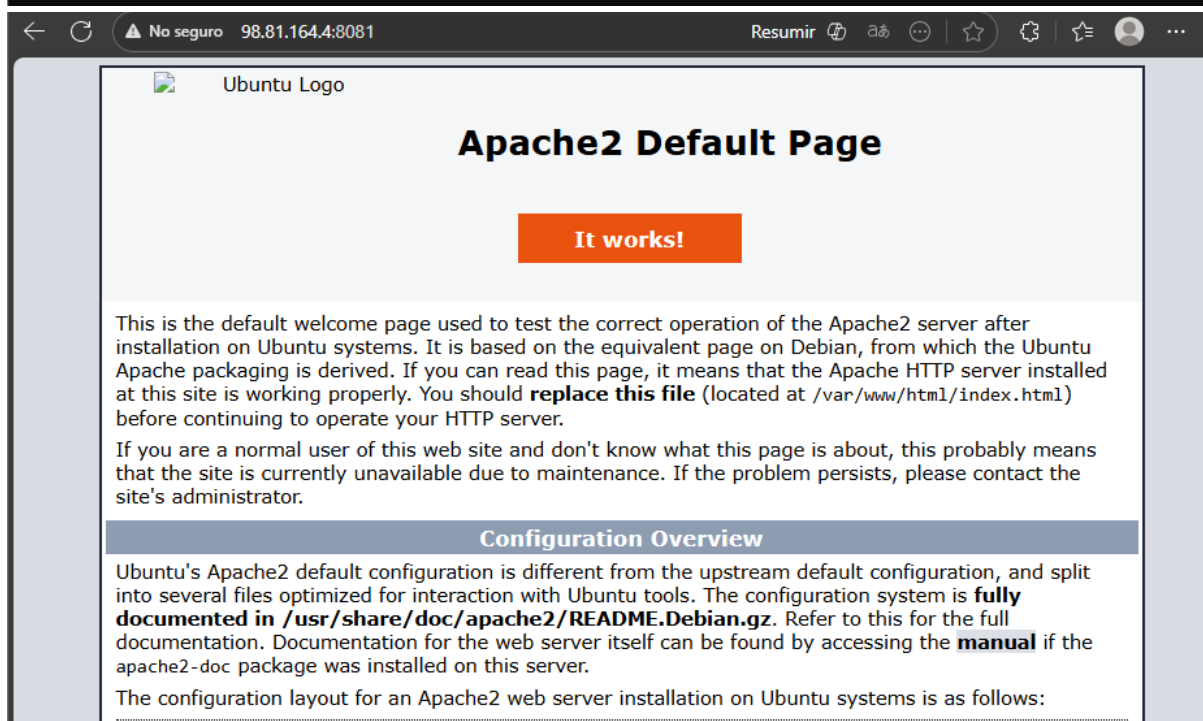
```
ubuntu@ip-172-31-23-219:~$ curl http://localhost:8081
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.d
td">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

      body, html {
        padding: 3px 3px 3px 3px;

        background-color: #D8DBE2;

        font-family: Ubuntu, Verdana, sans-serif;
        font-size: 11pt;
        text-align: center;
      }

      div.main_page {
        position: relative;
        display: table;
      }
    </style>
  </head>
  <body>
    <div class="main_page">
      <div style="text-align: center; margin-bottom: 20px;">
        <img alt="Ubuntu Logo" data-bbox="212 531 228 543" style="vertical-align: middle;"/>
        <h1 style="margin: 0; font-size: 2em; font-weight: normal;">Apache2 Default Page
        <div style="background-color: #f4a460; color: white; padding: 5px 15px; display: inline-block; margin-top: 10px;">It works!
      </div>
      <div style="text-align: left; margin-top: 20px;">
        <p>This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
        <p>If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
        <h3 style="background-color: #d8dbe2; padding: 5px; margin-top: 20px; text-align: center;">Configuration Overview
        <p>Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
        <p>The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
    </div>
    </div>
  </body>
</html>
```



PARTE 3: INSTALACIÓN Y CONFIGURACIÓN DE CADDY

1. Instalar dependencias necesarias

Comando:

```
sudo apt install -y debian-keyring debian-archive-keyring apt-transport-https curl
```

Descripción: Instala herramientas necesarias para añadir repositorios externos.

```
ubuntu@ip-172-31-23-219:~$ sudo apt install -y debian-keyring debian-archive-keyring apt-transport-https curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
The following NEW packages will be installed:
  apt-transport-https debian-archive-keyring debian-keyring
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 31.5 MB of archives.
After this operation, 33.4 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3970
  B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 debian-archive-keyring all 2023.4ubuntu1 [1
  68 kB]
```

2. Agregar repositorio de Caddy

Comando:

```
curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/gpg.key' | sudo gpg --dearmor -o
/usr/share/keyrings/caddy-stable-archive-keyring.gpg
```

```
ubuntu@ip-172-31-23-219:~$ curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/gpg.key' | sudo gpg --dearmor -o
/usr/share/keyrings/caddy-stable-archive-keyring.gpg
ubuntu@ip-172-31-23-219:~$ curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/debian.deb.txt' | sudo tee /etc/a
pt/sources.list.d/caddy-stable.list
# Source: Caddy
# Site: https://github.com/caddyserver/caddy
# Repository: Caddy / stable
# Description: Fast, multi-platform web server with automatic HTTPS

deb [signed-by=/usr/share/keyrings/caddy-stable-archive-keyring.gpg] https://dl.cloudsmith.io/public/caddy/stable/deb
/debian any-version main

deb-src [signed-by=/usr/share/keyrings/caddy-stable-archive-keyring.gpg] https://dl.cloudsmith.io/public/caddy/stable
/deb/debian any-version main
```

```
curl -1sLf 'https://dl.cloudsmith.io/public/caddy/stable/debian.deb.txt' | sudo tee
/etc/apt/sources.list.d/caddy-stable.list
```

Descripción: Añade el repositorio oficial de Caddy a tu sistema.

```
ubuntu@ip-172-31-23-219:~$ sudo apt update && sudo apt install caddy -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version InRelease [14.8 kB]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version/main amd64 Packages [4329 B]
Fetched 19.1 kB in 1s (32.4 kB/s)
```

5. Crear archivo Markdown de prueba.

Comando:

```
echo "# Bienvenido a Caddy" | sudo tee /var/www/caddy/README.md
```

```
echo "" | sudo tee -a /var/www/caddy/README.md
```

```
echo "Este servidor está funcionando correctamente." | sudo tee -a  
/var/www/caddy/README.md
```

```
echo "" | sudo tee -a /var/www/caddy/README.md
```

```
echo "## Características" | sudo tee -a /var/www/caddy/README.md
```

```
echo "- Servidor moderno" | sudo tee -a /var/www/caddy/README.md
```

```
echo "- HTTPS automático" | sudo tee -a /var/www/caddy/README.md
```

```
echo "- Fácil configuración" | sudo tee -a /var/www/caddy/README.md
```

```
ubuntu@ip-172-31-23-219:~$ sudo mkdir -p /var/www/caddy  
ubuntu@ip-172-31-23-219:~$ echo "# Bienvenido a Caddy" | sudo tee /var/www/caddy/README.md  
# Bienvenido a Caddy  
ubuntu@ip-172-31-23-219:~$ echo "" | sudo tee -a /var/www/caddy/README.md
```

```
ubuntu@ip-172-31-23-219:~$ echo "Este servidor está funcionando correctamente." | sudo tee -a /var/www/caddy/README.m  
d  
Este servidor está funcionando correctamente.  
ubuntu@ip-172-31-23-219:~$ echo "" | sudo tee -a /var/www/caddy/README.md
```

```
ubuntu@ip-172-31-23-219:~$ echo "## Características" | sudo tee -a /var/www/caddy/README.md  
## Características  
ubuntu@ip-172-31-23-219:~$ echo "- Servidor moderno" | sudo tee -a /var/www/caddy/README.md  
- Servidor moderno  
ubuntu@ip-172-31-23-219:~$ echo "- HTTPS automático" | sudo tee -a /var/www/caddy/README.md  
- HTTPS automático  
ubuntu@ip-172-31-23-219:~$ echo "- Fácil configuración" | sudo tee -a /var/www/caddy/README.md  
- Fácil configuración
```

6. Crear imagen de prueba (cuidado WSL hay que hacer ajustes)

Comando:

```
curl -o /tmp/test-image.jpg "https://www.python.org/static/apple-touch-icon-144x144-precomposed.png"
```

```
ubuntu@ip-172-31-23-219:~$ curl -o /tmp/test-image.jpg "https://www.python.org/static/apple-touch-icon-144x144-precomposed.png"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 7382  100 7382    0     0  161k    0 --:--:-- --:--:-- --:--:-- 163k
ubuntu@ip-172-31-23-219:~$ sudo mv /tmp/test-image.jpg /var/www/caddy/test.jpg
```

```
sudo mv /tmp/test-image.jpg /var/www/caddy/test.jpg
```

```
ubuntu@ip-172-31-23-219:~$ sudo mv /tmp/test-image.jpg /var/www/caddy/test.jpg
```

7. Crear Caddyfile personalizado

Comando:

```
sudo nano /etc/caddy/Caddyfile
```

```
ubuntu@ip-172-31-23-219:~$ sudo nano /etc/caddy/caddyfile
```

Descripción: Abre el archivo de configuración de Caddy.

Escribe el siguiente contenido:

```
:8082 {
```

```
root * /var/www/caddy
```

```
file_server
```

```
}
```

```
GNU nano 7.2 /etc/caddy/Caddyfile
# The Caddyfile is an easy way to configure your Caddy web server.
#
# Unless the file starts with a global options block, the first
# uncommented line is always the address of your site.
#
# To use your own domain name (with automatic HTTPS), first make
# sure your domain's A/AAAA DNS records are properly pointed to
# this machine's public IP, then replace ":80" below with your
# domain name.

:8082 {
    # Set this path to your site's directory.
    root * /var/www/caddy

    # Enable the static file server.
    file_server browse

    @markdown path *.md
    header @markdown Content-Type text/plain

    # Another common task is to set up a reverse proxy:
    # reverse_proxy localhost:8080

    # Or serve a PHP site through php-fpm:
    # php_fastcgi localhost:9000
}
```

8. Reiniciar Caddy

Comando:

```
sudo systemctl restart caddy
```

Descripción: Reinicia Caddy para aplicar la nueva configuración.

```
ubuntu@ip-172-31-23-219:~$ sudo systemctl restart caddy
ubuntu@ip-172-31-23-219:~$ sudo systemctl status caddy
● caddy.service - Caddy
   Loaded: loaded (/usr/lib/systemd/system/caddy.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-11-25 08:48:10 UTC; 6s ago
     Docs: https://caddyserver.com/docs/
   Main PID: 8231 (caddy)
    Tasks: 8 (limit: 1017)
   Memory: 47.9M (peak: 48.2M)
      CPU: 112ms
   CGroup: /system.slice/caddy.service
           └─8231 /usr/bin/caddy run --environ --config /etc/caddy/Caddyfile

Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"warn","ts":1764060490.4888577,"logger":"http.auto_https","ms
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"info","ts":1764060490.4894293,"logger":"tls.cache.maintenanc
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"warn","ts":1764060490.4900265,"logger":"http","msg":"HTTP/2 >
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"warn","ts":1764060490.4909449,"logger":"http","msg":"HTTP/3 >
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"info","ts":1764060490.49095,"logger":"http.log","msg":"serve
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"info","ts":1764060490.4944673,"msg":"autosaved config (load >
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"info","ts":1764060490.4945316,"msg":"serving initial configu
Nov 25 08:48:10 ip-172-31-23-219 systemd[1]: Started caddy.service - Caddy.
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"info","ts":1764060490.4999899,"logger":"tls","msg":"cleaning >
Nov 25 08:48:10 ip-172-31-23-219 caddy[8231]: {"level":"info","ts":1764060490.503197,"logger":"tls","msg":"finished >
```

9. Comando adicional para verificar puerto:

```
sudo netstat -tulpn | grep caddy
```

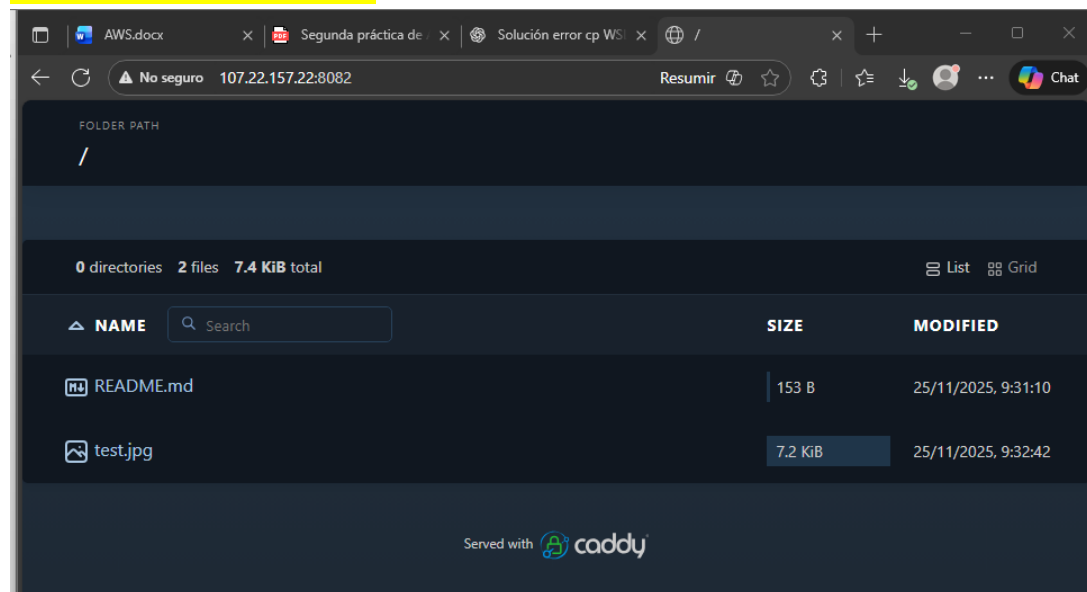
Descripción: Verifica en qué puerto está escuchando Caddy (debe mostrar 8082).

```
ubuntu@ip-172-31-23-219:~$ sudo netstat -tulpn | grep caddy
tcp        0      0 127.0.0.1:2019      0.0.0.0:*           LISTEN      1375/caddy
tcp6       0      0 :::8082             :::*                 LISTEN      1375/caddy
```

10. Probar Caddy desde terminal

Comando:

```
curl http://localhost:8082/
```

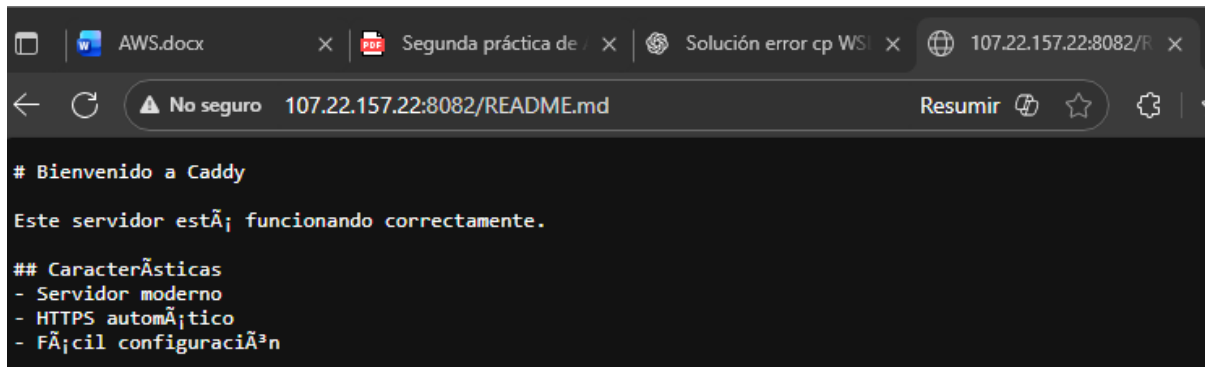


11. Probar archivo Markdown

Comando:

```
curl http://localhost:8082/README.md
```

Descripción: Verifica que Caddy sirve correctamente archivos Markdown.



PARTE 4: CONFIGURACIÓN DE HTTPS CON CERTBOT EN APACHE

1. Instalar Certbot y el plugin de Apache

Comando:

```
sudo apt install certbot python3-certbot-apache -y
```

Descripción: Instala Certbot y su integración con Apache para gestionar certificados SSL.

```
ubuntu@ip-172-31-23-219:~$ sudo apt install certbot python3-certbot-apache -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-aws-6.14-headers-6.14.0-1015 linux-aws-6.14-tools-6.14.0-1015 linux-headers-6.14.0-1015-aws
  linux-image-6.14.0-1015-aws linux-modules-6.14.0-1015-aws linux-tools-6.14.0-1015-aws
```

2. Verificar dominio o usar localhost

Nota: Para obtener certificados reales de Let's Encrypt necesitas un dominio público. Para esta práctica usaremos certificados autofirmados.

Comando:

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/apache-selfsigned.key -out /etc/ssl/certs/apache-selfsigned.crt
```

[illegible]

3. Habilitar módulo SSL en Apache

Comando:

```
sudo a2enmod ssl
```

Descripción: Activa el módulo SSL necesario para HTTPS en Apache.

```
ubuntu@ip-172-31-23-219:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
ubuntu@ip-172-31-23-219:~$ sudo systemctl restart apache2
ubuntu@ip-172-31-23-219:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
```

4. Cambiar puerto SSL

Comando:

```
sudo nano /etc/apache2/ports.conf
```

Descripción: Añade la línea Listen **8443** para que Apache escuche HTTPS en puerto **8443**.

```
GNU nano 7.2 /etc/apache2/ports.conf
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 8443_

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>
```


5. Modificar VirtualHost SSL

Comando:

```
sudo nano /etc/apache2/sites-available/default-ssl.conf
```

Descripción: Cambia `<VirtualHost *:443>` por `<VirtualHost *:8443>`.

```
GNU nano 7.2 /etc/apache2/sites-available/default-ssl.conf *
<VirtualHost *:8443>
    ServerAdmin webmaster@localhost
```

6. Habilitar sitio SSL

Comando:

```
sudo a2ensite default-ssl.conf
```

Descripción: Activa la configuración SSL en Apache.

```
ubuntu@ip-172-31-23-219:~$ sudo systemctl reload apache2
ubuntu@ip-172-31-23-219:~$ sudo a2ensite default-ssl.conf
Site default-ssl already enabled
```

7. Reiniciar Apache

Comando:

```
sudo systemctl restart apache2
```

Descripción: Aplica todos los cambios de configuración SSL.

```
ubuntu@ip-172-31-23-219:~$ sudo systemctl restart apache2
```

8. Verificar HTTPS

Comando:

```
curl -i -k https://localhost:8443
```

Descripción: Prueba la conexión HTTPS (el flag *-k* ignora el aviso del certificado autofirmado).

```
ubuntu@ip-172-31-23-219:~$ curl -i -k https://localhost:8443
HTTP/1.1 200 OK
Date: Thu, 27 Nov 2025 13:00:00 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Thu, 20 Nov 2025 13:28:48 GMT
ETag: "29af-64406ad4c60d1"
Accept-Ranges: bytes
Content-Length: 10671
Vary: Accept-Encoding
Content-Type: text/html

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.d
td">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

      body, html {
        padding: 3px 3px 3px 3px;

        background-color: #D8DBE2;

        font-family: Ubuntu, Verdana, sans-serif;
        font-size: 11pt;
        text-align: center;
      }

      div.main_page {
        position: relative;
        display: table;

        width: 800px;

        margin-bottom: 3px;
        margin-left: auto;
        margin-right: auto;
        padding: 0px 0px 0px 0px;
```



Ubuntu

Apache2 Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|
|   |-- ports.conf
|
|   |-- mods-enabled
|       |-- *.load
|       |-- *.conf
|
|   |-- conf-enabled
|       |-- *.conf
|
|   |-- sites-enabled
|       |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf`. See their respective man pages for detailed information.

PARTE 5: VERIFICACIÓN FINAL DE LOS TRES SERVIDORES

1. Verificar que todos los servicios están activos

Comando para verificar Apache:

```
sudo systemctl status apache2
```

Comando para verificar puerto de Apache:

```
sudo netstat -tulpn | grep apache2
```

Comando para verificar Nginx:

```
sudo systemctl status nginx
```

Comando para verificar puerto de Nginx:

```
sudo netstat -tulpn | grep nginx
```

Comando para verificar Caddy:

```
sudo systemctl status caddy
```

Comando para verificar puerto de Caddy:

```
sudo netstat -tulpn | grep caddy
```

```
ubuntu@ip-172-31-23-219:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-27 12:59:02 UTC; 6min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 1713 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 1717 (apache2)
    Tasks: 7 (limit: 1017)
   Memory: 13.1M (peak: 13.3M)
      CPU: 105ms
   CGroup: /system.slice/apache2.service
           └─1717 /usr/sbin/apache2 -k start
             └─1719 /usr/sbin/apache2 -k start
               └─1720 /usr/sbin/apache2 -k start
                 └─1721 /usr/sbin/apache2 -k start
                   └─1722 /usr/sbin/apache2 -k start
                     └─1723 /usr/sbin/apache2 -k start
                       └─1729 /usr/sbin/apache2 -k start
```

Nov 27 12:59:02 ip-172-31-23-219 systemd[1]: Starting apache2.service - The Apache HTTP Server..

Nov 27 12:59:02 ip-172-31-23-219 systemd[1]: Started apache2.service - The Apache HTTP Server.

```
ubuntu@ip-172-31-23-219:~$ sudo netstat -tulpn | grep apache2
tcp6      0      0 :::8080          :::*              LISTEN    13402/apache2
```

PARTE 5: VERIFICACIÓN FINAL DE LOS TRES SERVIDORES

1. Verificar que todos los servicios están activos

Comando para verificar Apache:

```
sudo systemctl status apache2
```

Comando para verificar puerto de Apache:

```
sudo netstat -tulpn | grep apache2
```

Comando para verificar Nginx:

```
sudo systemctl status nginx
```

Comando para verificar puerto de Nginx:

```
sudo netstat -tulpn | grep nginx
```

Comando para verificar Caddy:

```
sudo systemctl status caddy
```

Comando para verificar puerto de Caddy:

```
sudo netstat -tulpn | grep caddy
```

```
ubuntu@ip-172-31-23-219:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-27 13:07:58 UTC; 16s ago
     Docs: man:nginx(8)
  Main PID: 1758 (nginx)
    Tasks: 3 (limit: 1017)
   Memory: 3.7M (peak: 4.0M)
      CPU: 15ms
   CGroup: /system.slice/nginx.service
           └─1758 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1759 "nginx: worker process"
               └─1760 "nginx: worker process"
```

Nov 27 13:07:58 ip-172-31-23-219 systemd[1]: Starting nginx.service - A high performance web server and a reverse pr

Nov 27 13:07:58 ip-172-31-23-219 systemd[1]: Started nginx.service - A high performance web server and a reverse pro

lines 1-15/15 (END)

```
ubuntu@ip-172-31-23-219:~$ sudo netstat -tulpn | grep nginx
tcp        0      0 0.0.0.0:8081      0.0.0.0:*          LISTEN    1758/nginx: master
tcp6       0      0 :::80            :::*              LISTEN    1758/nginx: master
```

PARTE 5: VERIFICACIÓN FINAL DE LOS TRES SERVIDORES

1. Verificar que todos los servicios están activos

Comando para verificar Apache:

```
sudo systemctl status apache2
```

Comando para verificar puerto de Apache:

```
sudo netstat -tulpn | grep apache2
```

Comando para verificar Nginx:

```
sudo systemctl status nginx
```

Comando para verificar puerto de Nginx:

```
sudo netstat -tulpn | grep nginx
```

Comando para verificar Caddy:

```
sudo systemctl status caddy
```

Comando para verificar puerto de Caddy:

```
sudo netstat -tulpn | grep caddy
```

```
ubuntu@ip-172-31-23-219:~$ sudo systemctl status caddy
● caddy.service - Caddy
   Loaded: loaded (/usr/lib/systemd/system/caddy.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-27 12:47:29 UTC; 21min ago
     Docs: https://caddyserver.com/docs/
   Main PID: 526 (caddy)
    Tasks: 7 (limit: 1017)
   Memory: 48.0M (peak: 50.4M)
      CPU: 181ms
   CGroup: /system.slice/caddy.service
           └─526 /usr/bin/caddy run --environ --config /etc/caddy/Caddyfile

Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"info","ts":1764247649.733596,"logger":"admin","msg":"admin en>
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"info","ts":1764247649.737476,"logger":"tls.cache.maintenance">
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"warn","ts":1764247649.7389388,"logger":"http","msg":"HTTP/2 s>
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"warn","ts":1764247649.7389596,"logger":"http","msg":"HTTP/3 s>
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"info","ts":1764247649.738965,"logger":"http.log","msg":"serve>
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"info","ts":1764247649.740566,"msg":"autosaved config (load wi>
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"info","ts":1764247649.7406347,"msg":"serving initial configur>
Nov 27 12:47:29 ip-172-31-23-219 systemd[1]: Started caddy.service - Caddy.
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"info","ts":1764247649.7533104,"logger":"tls","msg":"cleaning >
Nov 27 12:47:29 ip-172-31-23-219 caddy[526]: {"level":"info","ts":1764247649.7554917,"logger":"tls","msg":"finished >

ubuntu@ip-172-31-23-219:~$ sudo netstat -tulpn | grep caddy
tcp        0      0 0.0.0.0:526         0.0.0.0:*           LISTEN      526/caddy
tcp6       0      0 :::526             :::*                 LISTEN      526/caddy
```

2. Verificar puertos en uso (todos simultáneamente)

Comando:

```
sudo netstat -tulpn | grep -E '8080|8081|8082|8443'
```

Descripción: Lista los cuatro puertos donde están escuchando los servidores simultáneamente.

```
ubuntu@ip-172-31-23-219:~$ sudo netstat -tulpn | grep -E '8080|8081|8082|8443'
tcp        0      0 0.0.0.0:8081        0.0.0.0:*           LISTEN      1758/nginx: master
tcp6       0      0 :::8082            :::*                 LISTEN      526/caddy
tcp6       0      0 :::8443            :::*                 LISTEN      1717/apache2
```

3. Probar todos los servidores

Comandos:

curl http://localhost:8080

curl http://localhost:8081

curl http://localhost:8082

curl -i -k https://localhost:8443



